FINANCIAL SYSTEMS RESEARCH LAB • ACADEMIC EVALUATION STUDY

Adaptive Temporal Graph Intelligence for Fraud Detection

Evaluating Relational Structure, Temporal Entity Memory, and Continual Adaptation on Streaming Transactions

PRESENTER A [Foundations & Design]
Problem Formulation • Causal Safeguards • Architecture
Heterogeneous Graph Construction • Temporal Memory Model

PRESENTER B [Implementation & Evidence]
Evaluation Protocol • Tabular vs Graph Findings • Ablations
Concept Drift Adaptation • Candidate Communities • Integrity

Rigorous validation-period empirical study on the IEEE-CIS Fraud Detection benchmark (590,540 transactions, 182-day span).

PART I — FOUNDATIONS & SYSTEM DESIGN • PRESENTER A

Transaction Fraud: Camouflage, Non-Stationarity, and Coordination

Why isolated transaction classification fails against sophisticated financial adversaries

Three Core Adversarial Dilemmas

1. Deceptive Camouflage
Individual fraudulent transactions deliberately mimic legitimate user spending in amounts, merchant categories, and purchase timing to bypass static rules.

2. Temporal Non-Stationarity & Drift
Fraud patterns evolve rapidly. Adversaries alter strategies across weeks, causing static detection models to suffer substantial decay in production.

3. Distributed Coordination
Organized rings distribute attacks across multiple cards, devices, and network proxies to keep velocity beneath account-level alert thresholds.

Benchmark Reality: The IEEE-CIS Dataset

590,540 Real-World Transactions
Spanning 182 continuous calendar days of payment activity.

Extreme Class Imbalance: 3.499% Positive Fraud
27.58 benign transactions for every single fraudulent transaction.

The Central Research Question:
Does the substantial engineering overhead of graph neural networks and dynamic entity memory yield defensible predictive gains over modern gradient boosting under strict chronological evaluation?

PART I — FOUNDATIONS & SYSTEM DESIGN • PRESENTER A

The Graph Illusion: Target Leakage and the Labelling Artifact

Why financial graph models often appear extraordinarily strong on paper while failing in production

THE 59.86% LABELLING ARTIFACT DISCOVERED IN IEEE-CIS

In IEEE-CIS, a single backward-looking feature—whether the same pseudo-account suffered earlier fraud—separates 59.86% fraud from 0.08%. Unconstrained graph convolutions accidentally propagate this labelling convention across neighbors or look backwards from future chargebacks, creating an illusion of high predictive power.

1. Strict Chronological Split

- Train: Days 1–121 (417,559 rows)
- Validation: Days 122–151 (83,655 rows)
- Held-Out: Days 152–182 (89,326 rows, cryptographically sealed)

Encoders, imputers, and scalers are fitted strictly on Train data only. Random k-fold splits are strictly rejected.

2. Causal Feature Masking

- Every feature for transaction t sees only strictly earlier rows (t' < t).
- Handcrafted graph features use historical entity counts with zero future leakage.
- Neural graphs score day t using graph state masked strictly to edges with timestamp <= t.

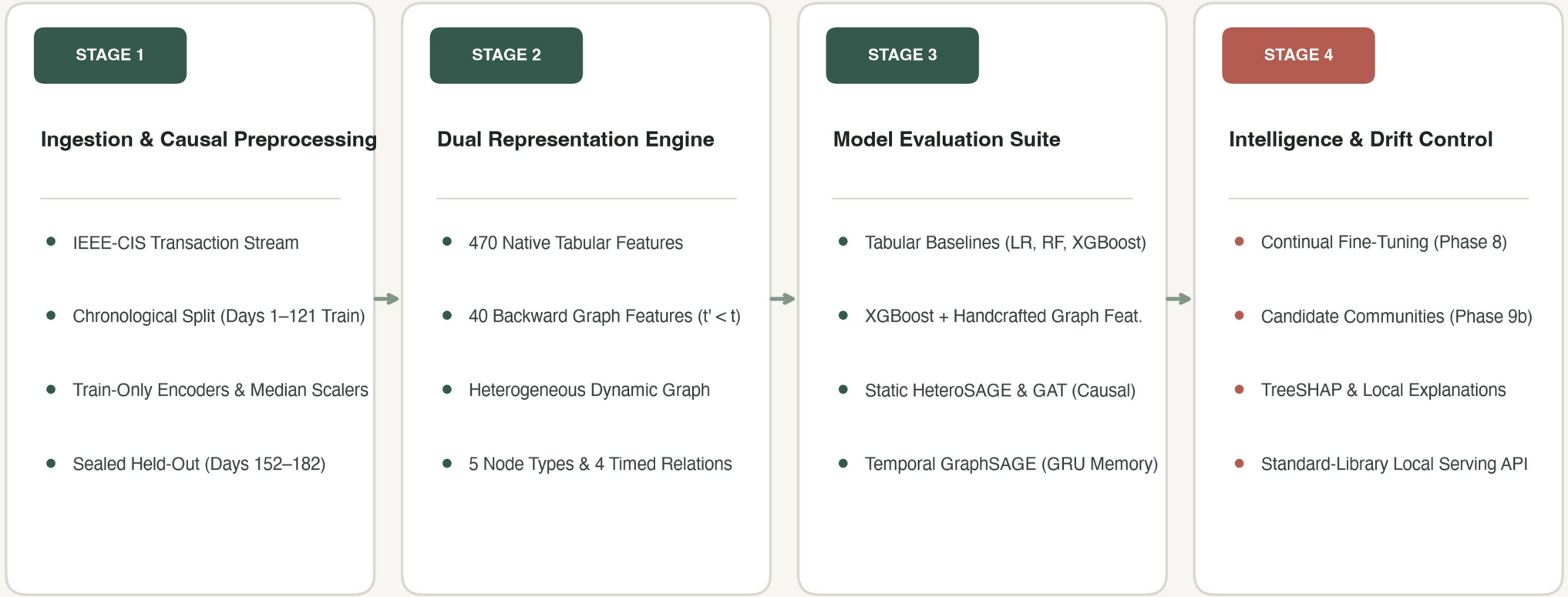
3. Interventional Checks

- Interventional code tests corrupt future days and verify that earlier predictions remain bit-identical.
- Complete separation between research evaluation and synthetic demonstration pipelines.
- Frozen validation gate architecture prevents post-hoc hyperparameter tuning.

PART I — FOUNDATIONS & SYSTEM DESIGN • PRESENTER A

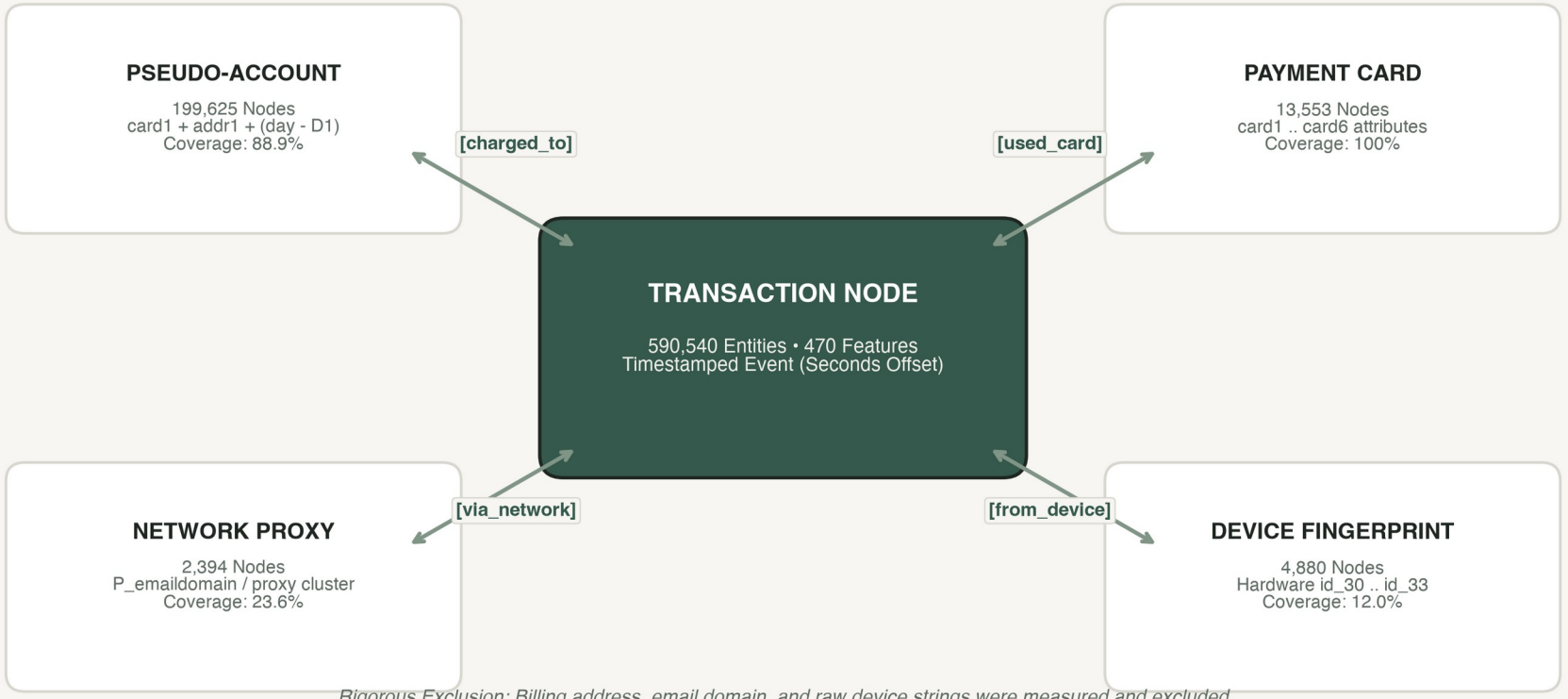
System Architecture: From Ingestion to Verified Intelligence

A dual-pipeline design separating tabular models, relational graphs, and temporal memory



Grounded Graph Construction: Entities and Timed Relational Edges

Constructing a heterogeneous network from observed transaction fields without speculative entities



Rigorous Exclusion: Billing address, email domain, and raw device strings were measured and excluded (produced massive hubs with zero predictive lift above random chance).

The Modeling Dilemma: Handcrafted Graph Features vs Deep GNNs

Two competing engineering paradigms for integrating relational context into tabular streams

Paradigm 1: Handcrafted Graph Features (B)

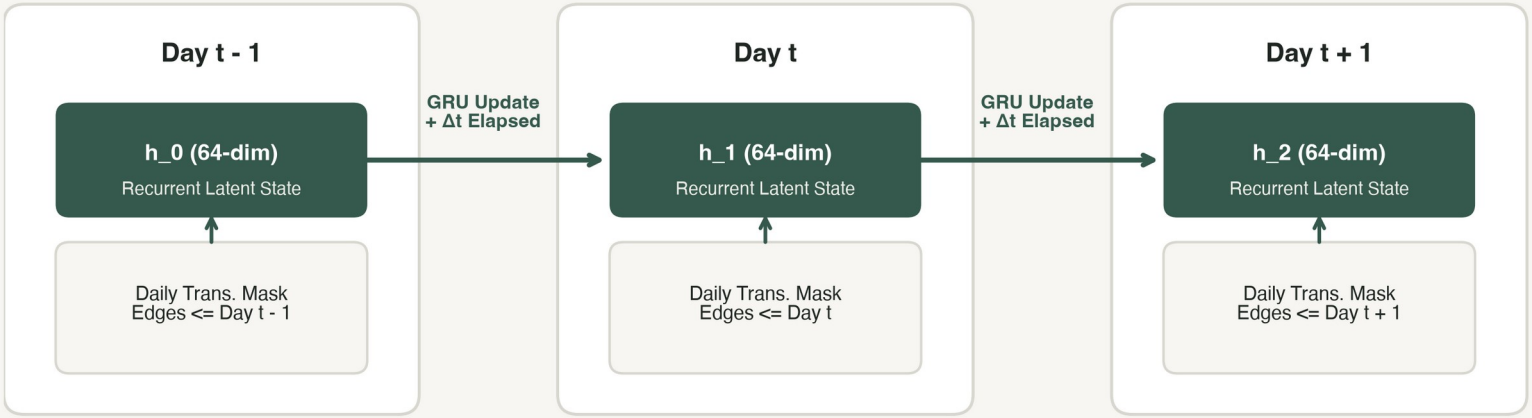
- **40 Label-Free Backward Features:** Computed per relation from strictly earlier transactions (t' < t) using running entity state.
- **Multi-Scale Temporal Velocity:** Transaction counts across prior 1-day and 7-day rolling windows for each associated entity.
- **Recency & Monetary Behavior:** Seconds elapsed since last transaction; running mean and standard deviation of log-transaction amount.
- **Relational Entity Diversity:** Count of distinct payment cards, devices, and network proxies linked to the pseudo-account to date.
- **Pre-Registered Protocol:** Feature formulas and acceptance rules locked in PREREGISTRATION_B.md before running any code.

Paradigm 2: Graph Neural Networks (C & D)

- **Heterogeneous GraphSAGE (Cell C):** Per-node-type linear projections with two relational SAGE aggregation layers (64 hidden units, 162k params).
- **Graph Attention Networks (Cell D):** Multi-head attention mechanism (4 heads, 364k params) designed to dynamically weight neighbor importance.
- **Causal Snapshot Masking:** GNNs score transactions on day t using only the graph structure formed by edges timestamped <= t.
- **No-Relations MLP Control:** An identical architecture with all relational edges removed (34k params) to isolate relational value from neural capacity.
- **Class-Weighted Optimization:** Trained with Adam (lr=1e-3) and class-weighted binary cross-entropy (weight=27.36) with early stopping.

Temporal GraphSAGE: Continuous Recurrent Entity State

Preserving dynamic behavioral memory across transactions without forward leakage



Strict Invariant: Entity states carry forward continuously without label leakage.
Predictions on day t use h_{t-1} and same-day structure without accessing future days or labels.

RIGOROUS EXPERIMENTAL CONTROLS (PHASE 7A)

To prove that any observed temporal gain is actually caused by the recurrent GRU mechanism rather than the training regime, we evaluated two pre-registered controls: (1) Schedule-Matched Static GNN (equal optimizer steps), and (2) Memory-Ablated Temporal GNN (replacing the recurrent GRU with a simple cumulative average).

Pre-Registered Research Hypotheses: What We Set Out to Test

Establishing rigorous falsification criteria before executing experimental runs

The Five Pre-Registered Research Hypotheses

- **H1: Relational Information Gain:** Graph structure provides predictive information beyond conventional transaction-level tabular features.
- **H2: Representation Superiority:** Learned graph neural representations (GNNs) outperform handcrafted backward-looking graph features.
- **H3: Temporal Memory Generalization:** Recurrent entity memory improves generalization to future transactions compared with static graph modeling.
- **H4: Continual Drift Adaptation:** Periodic fine-tuning on newly revealed labels significantly reduces model degradation under concept drift.
- **H5: Coordinated Community Discovery:** Graph-based community detection discovers coordinated multi-entity fraud structures that row-level classifiers miss.

PRESENTER HANDOFF

From System Architecture to Empirical Evidence

Presenter A has established the foundational problem, causal invariants, heterogeneous graph construction, and temporal architecture.

Presenter B will now present the experimental evaluation, our quantitative results across tabular and graph models, drift adaptation evidence, community discovery benchmarks, and final research conclusions.

PART II — IMPLEMENTATION & EVIDENCE • PRESENTER B

Experimental Protocol: Metrics, Segmentation, and Integrity

Why conventional accuracy metrics mislead and how we enforced verification rigor

1. PR-AUC as Primary Metric

Under 27.6:1 class imbalance, standard accuracy and ROC-AUC are deeply misleading:

- A naive classifier predicting 'legitimate' achieves 96.5% accuracy.
- ROC-AUC evaluates true positive rate against false positive rate over all negatives, yielding inflated scores (> 0.85) even when most alerts are false alarms.
- PR-AUC (Precision-Recall AUC) directly measures precision across recall thresholds and serves as our primary benchmark.

2. Strict 3-Way Segmentation

Aggregate metrics obscure where graph models succeed or fail. We report all results across 3 non-overlapping segments:

- Warm: The pseudo-account has prior transaction history.
- Cold-Start: Known entity, but this is its first observed transaction.
- No-Edge: Transaction with no linked pseudo-account.

Critical Rule: Merging Cold and No-Edge artificially inflates cold-start metrics and is strictly prohibited in our protocol.

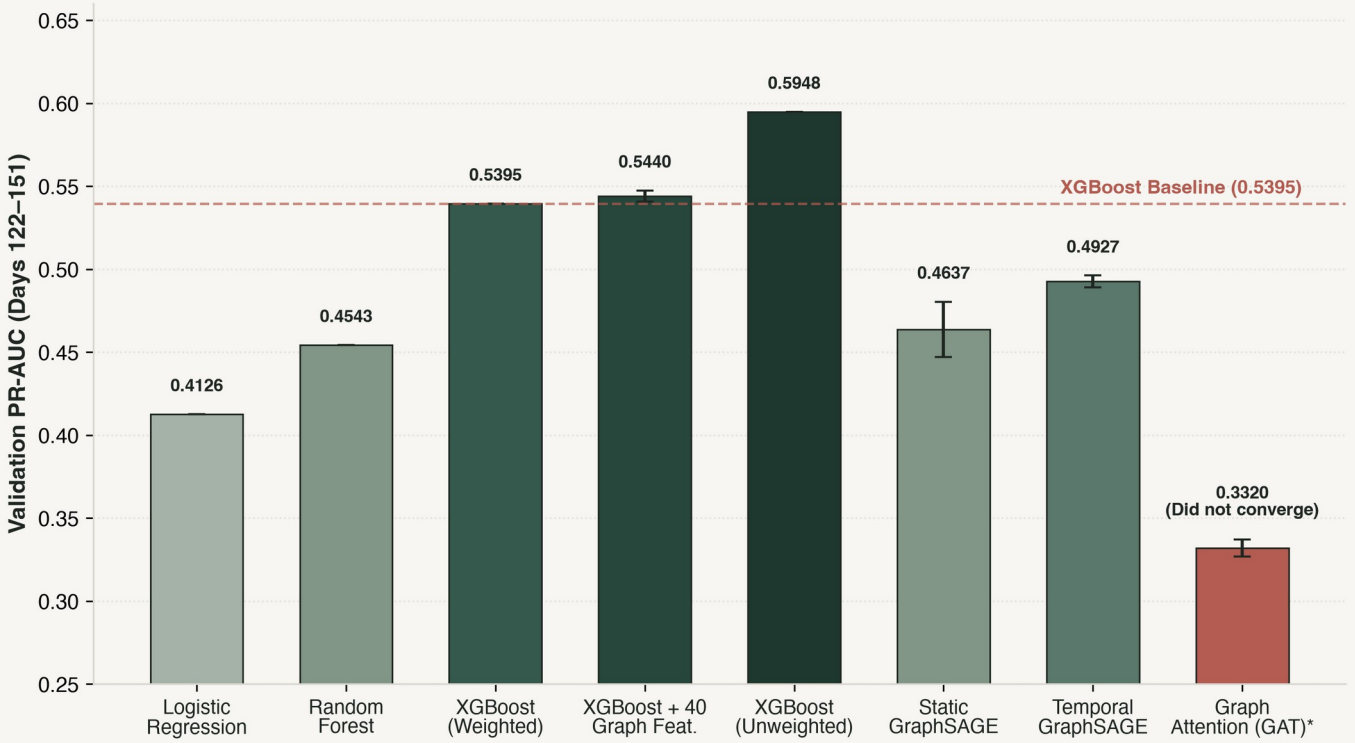
3. Verification & Sealed Test

All reported results are backed by cryptographic verification:

- Unified Metric Engine: A single shared module (src/models/metrics.py) recomputed 8,184 values across 65 files with exact 0.0 discrepancy.
- Sealed Held-Out Period: Days 152–182 remain locked under cryptographic guards. Zero held-out numbers are reported, eliminating post-hoc cherry-picking.
- All reported metrics reflect validation-period evaluation (Days 122–151).

The Tabular Baseline Prevails: Tuned XGBoost Outperforms Graph Models

Validation-period PR-AUC results (Days 122–151) across tabular, handcrafted, and neural architectures



Key Findings & Hypothesis Verdicts

- **XGBoost Baseline Strongest:** Tuned XGBoost on tabular features achieves 0.5395 PR-AUC (weighted) and 0.5948 (unweighted), outperforming all graph models.
- **Comparison B (XGB + 40 Graph Feat.):** Primary recipe reaches 0.5440 (+0.0046). Across 3 subsampled seeds, delta is -0.0005 vs SD 0.0034. H1-direct is NOT ESTABLISHED.
- **H2 Contradicted Under Rule:** Handcrafted graph features (0.5611) outperform Temporal GraphSAGE (0.4927) and GraphSAGE (0.4637). Learned embeddings did not beat handcrafted features.
- **GAT Convergence Failure:** GAT hit the 150-epoch ceiling at 0.3320 PR-AUC without converging. We report this transparently as an under-trained failure, not an inherent flaw.

PART II — IMPLEMENTATION & EVIDENCE • PRESENTER B

Temporal Memory Yields Modest, Cold-Start Concentrated Gains

Controlled ablation studies reveal where relational dynamics and memory actually matter

Temporal Memory Evaluation (H3)

- **Beat Memory-Ablated Control in 3 of 3 Seeds:** Temporal GraphSAGE (0.4927) outperforms the cumulative-mean control (0.4790) by +0.0137 pooled PR-AUC, exceeding the seed standard deviation of both arms.
- **Beat Schedule-Matched Static GNN in 2 of 3 Seeds:** Temporal model beats static GraphSAGE (0.4792) by +0.0135 on average, confirming that the gain is not an artifact of training schedule.
- **Gain Heavily Concentrated in Cold-Start:** On cold-start transactions (first transaction for a known entity), recurrent memory provides a notable +0.027 PR-AUC lift (0.3562 vs 0.3288).
- **Verdict on H3: Partially Supported:** Recurrent memory demonstrates a modest, statistically detectable benefit that is concentrated on early entity interactions.

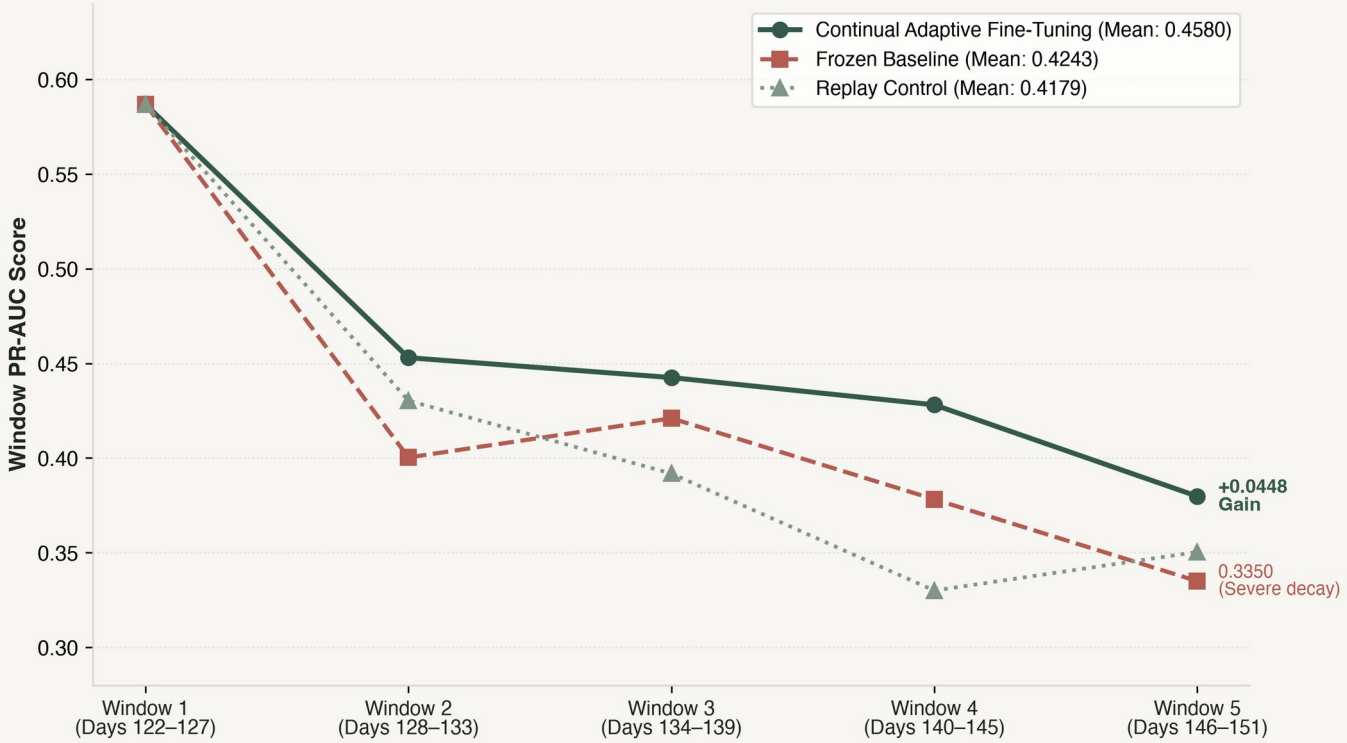
Relation Ablation Study (Temporal GraphSAGE)

- **Pseudo-Account ('charged_to') Is Critical:** Removing the pseudo-account relation drops pooled PR-AUC by -0.0309 across all 3 seeds (0.4618 vs 0.4927). In the warm segment, PR-AUC collapses from 0.3213 to 0.2290.
- **Device Fingerprint ('from_device') Contributes:** Removing device edges drops PR-AUC by -0.0073 in 3 of 3 seeds (0.4854 vs 0.4927), a modest contribution slightly above rerun noise.
- **Card ('used_card') & Network ('via_network') Non-Essential:** Removing used_card (-0.0038) or via_network (-0.0053) changes PR-AUC within our 0.0058 run-to-run noise floor. Removing them does not hurt performance.
- **Key Structural Takeaway:** Nearly all relational signal exploited by the temporal model flows through the derived pseudo-account identity key.

PART II — IMPLEMENTATION & EVIDENCE • PRESENTER B

Adapting Under Drift: Periodic Fine-Tuning Outperforms Static Models

Evaluating walk-forward adaptation across five consecutive 6-day validation windows



Drift Dynamics & H4 Verdict

- **Severe Performance Decay:** Frozen model PR-AUC collapses from 0.5868 in Window 1 to 0.3350 in Window 5 (a drop of > 0.25) over just 30 calendar days.
- **Mild Covariate Shift:** Feature PSI across 470 features averages only ~0.01. Input distributions barely moved, yet predictive efficacy decayed sharply.
- **H4 Supported Under Locked Rule:** Adaptive arm (0.4580) degrades significantly less than Frozen (0.4243) and beats Replay Control (0.4179) in 3 of 3 seeds (+0.0402 margin).
- **Replay Control Proves Cause:** Replay control resampled old training data with equal compute. It scored below Frozen, proving that recent labels—not compute volume—drive the gain.
- **Triggered Refresh Failure:** A score-distribution PSI trigger never fired (PSI <= 0.025 vs 0.10 threshold) while accuracy plunged. Label-free output drift failed to signal decay.

PART II — IMPLEMENTATION & EVIDENCE • PRESENTER B

Coordinated Community Discovery and Transparent Explanations

Uncovering multi-entity structures and generating faithful model attributions

Candidate Community Discovery (H5)

- **Bipartite Louvain Modularity Clustering:** Constructed dynamic bipartite graphs linking card tokens to shared infrastructure (devices, netproxies) over 7-day rolling windows.
- **Original Synthetic Benchmark Failed Gate G4:** In Phase 9, the original synthetic benchmark failed frozen interference gate G4 on 2 of 3 seeds. Following scientific integrity rules, H5 was declared NOT ESTABLISHED.
- **Phase 9b Pre-Registered Benchmark Passed:** A separately pre-registered benchmark with corrected interference ranking passed all gates. The structure-plus-time method recovered camouflaged planted groups (AUPRC 0.14–0.27 vs <0.05 baselines).
- **Essential Terminology Boundary:** Real-data clusters are strictly designated 'candidate coordinated communities'. Because IEEE-CIS contains zero verified fraud ring ground truth, claiming confirmed fraud rings is scientifically false.

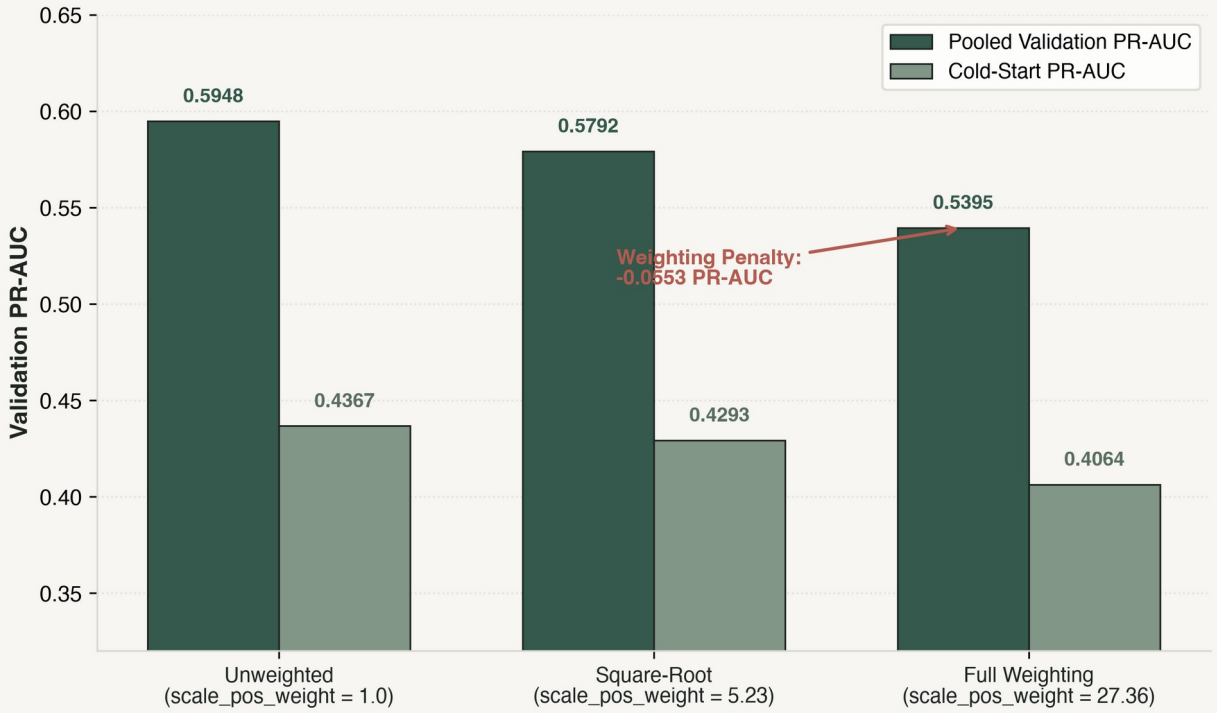
Transparent Explanations & Evidence

- **Exact TreeSHAP Feature Attributions:** For XGBoost, native TreeSHAP feature contributions sum exactly to the model margin (verified error < 8.1e-06 across 300 pre-registered validation rows).
- **Relational Feature Influence in Model B:** In Comparison B, graph features appear in the top-10 contributions for 55.3% of transactions (40.0% of top-scoring alerts and 70.7% of uniform draws).
- **Counterfactual Link Removals in GNNs:** For Temporal GraphSAGE, link removal perturbations confirmed pseudo-account edge removal changed output logit by 3.158 on average.
- **Attribution vs Causality Disclaimer:** SHAP values and link sensitivities faithfully describe model margin sensitivity, but do not establish real-world real-time causal guilt.

PART II — IMPLEMENTATION & EVIDENCE • PRESENTER B

The Class Weighting Penalty and Production-Grade Software

Surprising loss dynamics and an un-mocked standard-library serving stack



Production Software Implementation

- **The Class Imbalance Discovery:** Common practice applies full class weighting (scale_pos_weight = 27.36). Unweighted XGBoost (0.5948) beats full weighting (0.5395) by +0.0553 PR-AUC! Weighting inflates false alarms at high score thresholds—a swing larger than any graph enhancement.
- **Zero-Dependency Local Serving API:** Built entirely using Python's standard library (http.server, socketserver). Serves recorded validation tables, aggregate cards, and live scoring endpoints without Flask, FastAPI, or Node.
- **Working End-to-End Synthetic Demo:** scripts/demo_end_to_end.py runs on any machine with zero data prerequisites. Executes causal feature extraction, XGBoost scoring, and Louvain community discovery with byte-identical output.
- **381 Automated Tests with Mutation Testing:** Full test suite verifying leakage invariants, serving contracts, and error responses. Mutation testing confirmed 15 of 17 test mutants detected.

Conclusions: Lessons for Financial Graph Machine Learning

Three core takeaways, acknowledged limitations, and future research directions

Three Core Scientific Takeaways

1. The Enduring Power of Tabular Baselines

A tuned gradient-boosted tree on transaction features remains exceptionally difficult to beat. Relational graph intelligence must be evaluated against strong tabular baselines under rigorous causal controls.

2. Specificity of Relational & Temporal Value

Graph and temporal dynamics provide genuine value, but in targeted niches: recurrent memory delivers meaningful lift (+0.027 PR-AUC) in cold-start scenarios, and periodic adaptation successfully counters drift.

3. Scientific Hygiene Over Sensation

Negative findings, failed benchmark gates, and sealed held-out periods are essential research contributions that prevent the reproducibility crises prevalent in applied financial ML.

Honest Limitations & Next Steps

- **Validation-Period Scope:** All empirical claims reflect the validation period (Days 122–151). The held-out period (Days 152–182) remains cryptographically sealed awaiting authorization.
- **GNN Batch Latency vs Streaming Serving:** Full-batch GNNs require substantial GPU memory and whole-day graph state, making them impractical for single-row sub-100ms real-time payment authorization.
- **Candidate Communities vs Fraud Rings:** Unsupervised graph clusters reflect behavioral similarity; formal operational fraud-ring confirmation requires forensic ground truth unavailable in public data.
- **Future Work: Streaming Graph Stores:** Developing incremental dynamic graph representations that maintain cold-start entity memory within millisecond production latencies.